

MA 571: Introduction to Approximation Algorithms

Course Code	MA 571
Course Name	Introduction to Approximation Algorithms
L-T-P-C	3-1-0-4
Prerequisites	Data Structures and Algorithms, Linear Algebra, Probability
Intended for	B.Tech, M.Tech/MSc, Ph.D.
Distribution	Elective
Approval	59th BoA (Apr 2025)

Course Contents

- **Introduction:** [6 Lectures]
 - Why do we need approximation algorithms?
 - The Class P, The Class NP, Cook Reductions, Karp Reductions, and
 - NP-Completeness, coNP.
 - Fundamentals and Definitions Pertaining to Approximation Algorithms.
- **Greedy Approximation Algorithms:** Vertex Cover, Set Cover, Layering, k-center clustering problem, Shortest superstring via set cover, Facility location problems. [6 Lectures]
- **Combinatorial Algorithms:** Traveling Salesman Problem, Hardness of Approximation of TSP, Metric TSP, Steiner Tree, Minimum Makespan Scheduling, Bin Packing, PTAS for Bin Packing, Multiway Cut and k-Cut. [10 Lectures]
- **Dynamic Programming Algorithms:** [4 Lectures]
 - Why do we need approximation al
 - A pseudo-polynomial time algorithm for knapsack.
 - An FPTAS for knapsack, Subset Sum Problem.
- **Local Search:** Finding minimum-degree spanning trees, Edge coloring, Max-Cut. [4 Lectures]
- **Randomized Algorithms:** Max 3-SAT, Min Cut Algorithm. [3 Lectures]
- **Linear Programming (LP) Based Algorithms:** Vertex Cover via LP, Set Cover via LP, The primal-dual method. [5 Lectures]
- **Hardness of Approximation:** [4 Lectures]
 - Techniques in proving the hardness of approximation.
 - Approximation preserving reductions.

Text Books

1. Vazirani, Vijay V., **Approximation Algorithms**, Vol. 1, Springer, 2001.
2. Williamson, David P., Shmoys, David B., **The Design of Approximation Algorithms**, Cambridge University Press, 2011.

References

1. Hochbaum, Dorit S., **Approximation Algorithms for NP-hard Problems**, PWS Publishing Company, 1995.
2. Har-Peled, Sarel, **Geometric Approximation Algorithms**, American Mathematical Society, 2011.
3. Ausiello, Giorgio, et al., **Complexity and Approximation: Combinatorial Optimization Problems and Their Approximability Properties**, Springer Science & Business Media.